Status	Finished
Started	Sunday, 25 October 2026, 12:35 PM
Completed	Sunday, 25 October 2026, 12:38 PM
Duration	3 mins 29 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

- a. Dividing the input into parts, solving each part, and combining the solutions
- b. Sorting data sequentially
- c. Sorting data in a single pass
- d. Rearranging data without sorting

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **2**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

- a. Selection
- b. Distribution
- c. Insertion
- d. Extraction

The correct answer is:
Insertion

Question **3**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

a. By comparing elements sequentially
b. By breaking the input into smaller parts and merging them
c. By sorting data in a single pass
d. By using the bubble-up method

The correct answer is: By breaking the input into smaller parts and merging them

Question **4**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

a. Linear Search
b. Merge Sort
c. Bubble Sort
d. Quick Sort

The correct answer is: Merge Sort

Question **5**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

a. Quick sort
b. Heap sort
c. Merge sort
d. Selection sort

The correct answer is:
Merge sort

Question **6**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

a. Bubbling up the smallest element
b. Bubbling up the largest element to its correct position
c. Dividing the list into halves
d. Sorting the entire list in one pass

The correct answer is: Bubbling up the largest element to its correct position

Question **7**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

a. It allows for quicker identification of duplicates
b. It complicates the analysis process
c. It has no effect on the analysis process
d. It makes the analysis slower

The correct answer is: It allows for quicker identification of duplicates

Question **8**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

a. Finding a specific element in a list
b. Arranging data in a specific order
c. Merging two datasets
d. Removing duplicates from a list

The correct answer is: Arranging data in a specific order

Question **9**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

- a. sort()
- b. order()
- c. arrange()
- d. sorted()

The correct answer is: sort()

Question **10**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

- a. It does not require recursion
- b. It is less efficient than bubble sort
- c. It sorts data using a single pass
- d. It is based on the divide-and-conquer approach

The correct answer is: It is based on the divide-and-conquer approach

Question **11**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

- a. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys
- b. A sorting algorithm is stable if it preserves the order of all keys
- c. A sorting algorithm is stable if it preserves the order of non-duplicate keys
- d. A sorting algorithm is stable if it preserves the order of duplicate keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **12**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

- a. It sorts data out of the box efficiently
- b. It only works with integer arrays
- c. It requires external libraries
- d. It is less efficient than custom sorting algorithms

The correct answer is: It sorts data out of the box efficiently

Question **13**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

- a. All of the above
- b. Duplicates
- c. Searching
- d. Selection

The correct answer is: All of the above

Question **14**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

- a. Linear Search
- b. Merge Sort
- c. Bubble Sort
- d. Binary Search

The correct answer is: Bubble Sort

Question **15**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

- a. Divide and conquer
- b. Backtracking
- c. Dynamic programming
- d. Greedy method

The correct answer is:
Divide and conquer